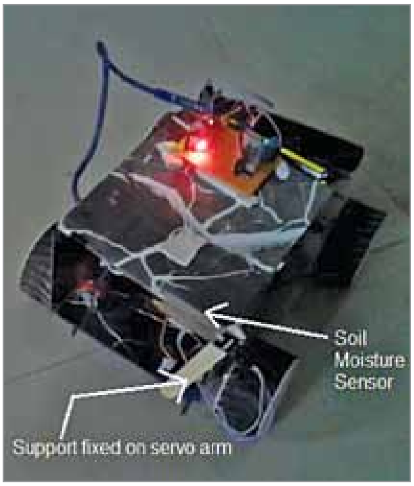


*Fig. 3: Author's prototype*


*Fig. 5: Dry soil status*


*Fig. 4: Various sensors' data captured on mobile phone*

To move the robot in any direction, command is required from smartphone through Bluetooth based Android application, called Bluetooth Terminal HC-05 by mightyIT. To do this, open (start) Bluetooth Android application in smartphone and search for HC-05 Bluetooth module. Once the phone detects HC-05 module, pair it with the smartphone's Bluetooth app. Enter passkey 1234 or 0000 the first time to pair with HC-05 module.

Now you can send commands from the smartphone to robot to move using the app. Following commands given in the table are used to move the robot (all these commands are already set in Android application):

| Command | Robot movement |
|---|---|
| 'F' | Move forward |
| 'R' | Turn right |
| 'L' | Turn left |
| 'B' | Move backward |
| 'S' | Stop moving |

When any of the above commands is sent (by sending the character in capital letter), it is received by HC-05 module. The module further gives this command to Arduino serially through its Tx-Rx pins. Arduino gets this command and compares it with set commands. If they match, the robot moves in the desired direction.

Once robot is in motion, it keeps moving (within the range) until command 'S' is sent to stop it. When robot stops, it moves the servo motor by 90° so that soil moisture sensor can go down into the soil to capture soil moisture value. At the same time, it starts reading sensor values from DHT11 and LDR. It reads analogue voltage output from soil moisture sensor and LDR and converts it to the range of 0-100%. It also reads digital values of temperature and humidity from DHT11 sensor.

Robot transmits all four values of these sensors to the smartphone through Bluetooth module. It keeps transmitting these values every three seconds till it is stopped by pressing 'S' command. When any of the commands (F,B,R,L) is given, the corresponding servo motor moves from 90° to 0° and the soil moisture sensor moves up. The robot stops transmitting sensor readings and starts moving. Thus it gives an idea of ambient temperature, humidity, soil moisture, and light intensity in the area.

Operation of the circuit is controlled by the program embedded in Arduino Uno microcontroller ATMega328.

## Software

The program code (BT_controlled_robot.ino) is written in Arduino programming language. It was tested using Arduino IDE version 1.8.18. Before compiling and uploading the code, make sure you include the relevant libraries, such as